

Alexandre Autran

Student in mathematics and machine learning

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Profile

I am a machine learning and mathematics student at the École Normale Supérieure (ENS) in Paris-Saclay and Rennes. My research focuses on probabilistic and geometric methods for machine learning, stochastic processes and stochastic control, high-dimensional statistics, density analysis, optimal transport, graph-based learning, and statistical learning theory.

Education

2024–2027 **ENS Diploma in Mathematics - Normalien**

Admitted to the ENS through a competitive entrance examination.

2026–2027 **Research year in Machine Learning**, *École Normale Supérieure Paris-Saclay*, Paris, France

Final year of the ENS Diploma (research track at ENS de Rennes).

2025–2027 **MSc in Mathematics**, *École Normale Supérieure de Rennes*, France

MSc in Mathematics (M2). Highest honors (17/20), first semester.

MSc in Mathematics (M1). Highest honors (17.6/20). Rank: 5th.

2024–2025 **BSc in Mathematics, Computer Science minor**, *École Normale Supérieure de Rennes*, France

High honors (15.4/20).

Publications

- [4] **A. Autran**, “Improved Wasserstein Estimates for Jump-Diffusion Perturbations under Semigroup Smoothing”, preprint, 2026.
- [3] **A. Autran**, “Second-Order Asymptotics of Emergent α -Stable Common Noise”, preprint, 2026.
- [2] **A. Autran**, “Optimization with an Exact Discrete Adjoint for a Diffusive Lotka–Volterra Harvesting Problem”, preprint, 2026.
- [1] **A. Autran**, D. Munteanu, and J.-L. Autran, “Multi-Dimensional Charge Diffusion–Collection Model in Semiconductor Devices Subjected to Single Ionizing Particles”, *Applied Sciences*, vol. 16, no. 11, Art. no. 5551, 2026. [[Paper](#)]

Research Experience

Aug 2026– **Machine Learning Researcher**

High-Dimensional Density Analysis Using Local Unimodality.

Supervised by Argyris Kalogeratos (Senior Researcher) at Centre Borelli, ENS Paris-Saclay, France.

May–Aug 2026 **MSc Research Internship**

Optical Modeling of Thermochromic VO₂-Based Oxide Nanocomposites [[Report](#)] [[Presentation](#)]

Supervised by Prof. Nguyen Ngoc Duy and Amaury Baret (PhD Student) at the University of Liège, Belgium.

Apr–Jun 2026 **MSc Research Internship**

Numerical Simulation and Optimal Control of Spatial Lotka–Volterra Systems. [[Report](#)]

Supervised by Prof. Martin J. Gander and Bastien Chaudet-Dumas (Postdoctoral Researcher) at the University of Geneva, Switzerland.

Apr–May 2026 **Machine Learning Projects**

Cardiovascular Disease Prediction Using Statistical Learning and Neural Network Models; Air Pollution Prediction from Environmental Data. [[Cardiovascular report](#)] [[Air-pollution report](#)]

Studies carried out as part of the MSc course *Statistical Learning* (Prof. Myriam Maumy).

Jan 2026 **Statistical Reading Group**

Supervised Statistical Learning – Prediction by Empirical Risk Minimization. [[Presentation](#)]

Supervised by Prof. Magalie Fromont at ENS de Rennes, France.

- Nov–Dec 2025 **MSc Research Seminar**
Point Process and Stein's Method. [[Report](#)] [[Presentation](#)]
Supervised by Prof. Jean-Christophe Breton at the University of Rennes, France.
- Oct–Nov 2025 **Algebra Reading Group**
Elimination Theory and Extension. [[Presentation](#)]
Supervised by Jade Nardi (Research Scientist) at *ENS de Rennes*, France.
- Jun–Aug 2025 **BSc Research Internship**
Lattice-Based Post-Quantum Cryptography. [[Report](#)] [[Presentation](#)]
Supervised by Prof. Pierre-Alain Fouque and Aurore Guillevis (Research Scientist) at Inria, France.
- May–Jun 2025 **BSc Research Internship**
Algebraic Counting of the Real Roots of Polynomials. [[Report](#)] [[Presentation](#)]
Supervised by Prof. Marie-Françoise Roy at the University of Rennes, France.
- Feb–Mar 2025 **Directed Research Reading**
Geometry of Numbers. [[Report](#)] [[Presentation](#)]
Supervised by Juliette Bavard (Research Scientist) at ENS de Rennes, France.
- Feb–Mar 2024 **BSc Research Internship**
Topological Study of Matrix Groups. [[Report](#)]
Supervised by Prof. Françoise Dal'Bo and Jérémy Bettinger (PhD Student) at the University of Rennes, France.
- May–Jun 2023 **BSc Research Internship**
Introduction to Hyperbolic Geometry and Fuchsian Groups. [[Report](#)] [[Presentation](#)]
Supervised by Frank Loray (CNRS Research Director) at the University of Rennes, France.

Other Activities

- 2024–2025 **ENS Oral Examinations**
Supervision of 2nd year BSc students in preparation for the oral entrance examinations in mathematics of the Écoles Normales Supérieures: Ulm, Paris-Saclay, Rennes and Lyon. [[Resources](#)]
- 2022–2025 **Teaching in Mathematics**
Tutoring for final-year high school students of the Lycée Chateaubriand in Rennes, and for first and second years BSc students of the University of Rennes. [[Resources](#)]
- 2023–2024 **BSc Student Representative**
Representative on the council of the Mathematics Department of the University of Rennes.
- Before 2022 **High School Competitions**
Concours Général in Mathematics, Physics, Philosophy, and National High School Mathematics Olympiad at Lycée Thiers in Marseille.

Computer Skills

Programming: Python, C/C++, Java, MATLAB, R.
Python / Machine Learning: NumPy, pandas, scikit-learn, PyTorch.

Languages

French: native language. **English:** professional working proficiency.

Sports and Interests

Sports: running, cycling, strength training.
Interests: reading and writing, in fantasy and science fiction literature; chess.

Academic Reference

François Bolley [[Web page](#)]
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